Benchmarking X: A Reproducible Evaluation Protocol for Y

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Abstract

This paper introduces a benchmark and evaluation protocol aimed at reliably measuring Y. We describe the design principles, task definitions, metrics, and validation checks that ensure the benchmark is diagnostic, reproducible, and challenging. Evaluations of representative systems illustrate capability gaps and variability across data slices, highlighting where current practice may overestimate performance. The contribution is an open, carefully validated evaluation setup that enables more informative assessment rather than a new learning method.

1. Introduction

Reliable progress on Y requires evaluations that go beyond single-number scores on narrow test sets. Many current protocols overfit to a fixed distribution, under-report uncertainty, and offer little diagnostic value when models fail under realistic shift. Our benchmark targets systematic measurement of Y across domains and demographic or topical slices, with contamination resistance and strict reproducibility. Design principles include coverage and realism, difficulty calibration, slice-wise reporting, and metric robustness. Contributions: a versioned benchmark and protocol; a taxonomy of task variants and difficulty bands; validated metrics with calibrated confidence intervals; and empirical findings showing sizable gaps between in-domain and shifted settings. Concretely, we frame Y as a family of tasks with shared inputs and distinct nuisance factors; the benchmark asks whether systems maintain accuracy, calibration, and ranking consistency across slices, not only the average score. Artifacts are versioned and hashed, and each release ships a manifest that pins data, scorer, and seeds so results can be exactly reproduced.

2. Related Work

Prior suites emphasize scale or realism but often lack slice-level diagnostics or leakage controls, e.g., FaceForensics++ [1], DFDC [2], and DF40 [3]. Foundation-model evaluations highlight transfer but rarely quantify calibration and group disparities [4]. Our work complements these by standardizing decontamination, reporting protocols, and uncertainty estimation for Y.

3. Benchmark Design

We define target capabilities, task formulations, input-output formats, and the evaluation setting. Tasks cover binary and multi-class Y with short- and long-form inputs; outputs are normalized to a common JSON record for consistent scoring. We specify training/validation/test splits with a frozen hidden test served via an evaluation server with signed submissions. Decontamination uses 5-gram and embedding search. Metrics include macro-F1, AUROC, and ECE with bootstrap intervals and calibrated CIs.

Taxonomy spans domains, lengths, and nuisance factors. Sampling is stratified to preserve marginals and enable slice evaluation. We ship a reference scorer, canonical prompts, and a JSON schema; submissions are client-side validated and signed. Each instance stores metadata (domain, length, nuisance tags, and a source hash); difficulty bands come from blinded pilot studies and remain fixed across versions. The evaluation server validates schema, verifies hashes, and rate-limits models to reduce test adaptation.

4. Benchmark Validation

We report annotation reliability, distributional diversity, difficulty calibration, decontamination and leakage checks, and sanity tests showing the benchmark is neither trivial nor broken. We measure Krippendorff's alpha, slice balance, and metric sensitivity to show conclusions are stable across reasonable choices; we probe leakage against public corpora using both lexical and semantic search. We further probe metric monotonicity and ranking stability under resampling; bootstrap intervals and permutation tests indicate robustness to seeds and splits.

We validate task clarity via expert spot checks, test rank agreement between metrics and experts, and stress-test under label noise and class imbalance. A negative-control split (random labels) confirms metrics avoid spurious gains;

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leakage probes remain stable across dedup thresholds.

5. Experiments

We evaluate strong baselines (heuristics, classical models, and zero-/few-shot foundation models [4]) under fixed seeds and compute 95% bootstrap confidence intervals. We include majority and random baselines and focus on what the benchmark reveals about strengths, weaknesses, and brittleness. We release prompts, hyperparameters, and seed files; all results report mean$\pm$CI across three runs for reproducibility.

Systems run under fixed decoding and compute budgets; prompts and hyperparameters are frozen per variant. To limit test adaptation, we cap submissions per model per week and require prediction provenance. We report overall and slice scores with stratified bootstrap CIs.

6. Analysis

We provide diagnostic analyses across slices, difficulty levels, and perturbations, including agreement with expert judgment where relevant. Figure 3 summarizes robustness patterns under controlled perturbations.

Errors concentrate on long inputs, cross-domain transfer, and nuisance perturbations; calibration degrades on rare slices and top-model ranking is unstable under mild shifts.

7. Discussion

The benchmark should be used to assess Y within the covered domains and tasks; ecological validity and residual contamination risk remain limitations. The results inform where systems are reliable versus brittle, but do not imply broader general intelligence or safety beyond this scope.

8. Conclusion

We summarize the evaluation contribution and key empirical insights enabled by the benchmark.

References

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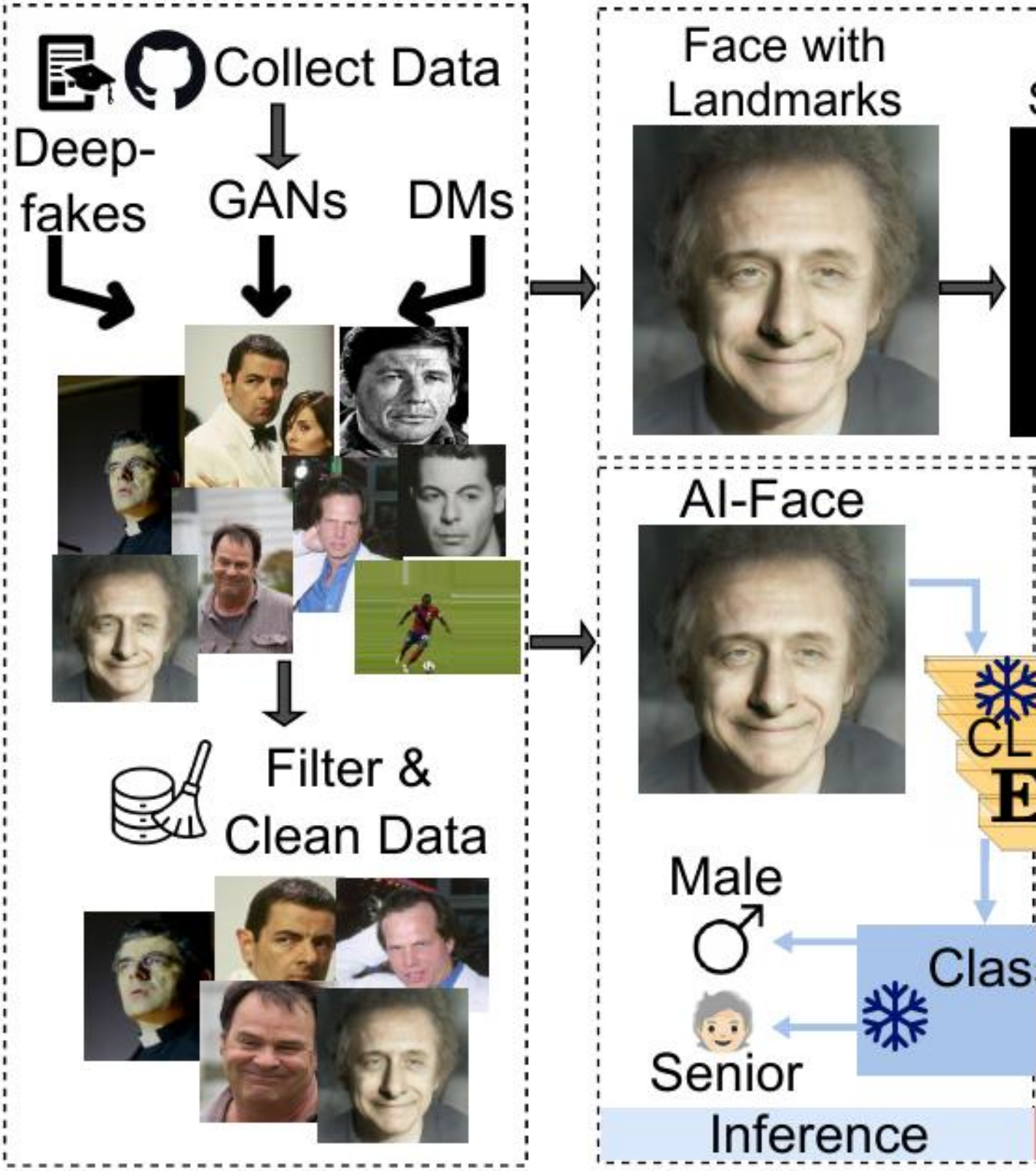

Figure 1. Benchmark overview: tasks, slices, metrics, and protocol.

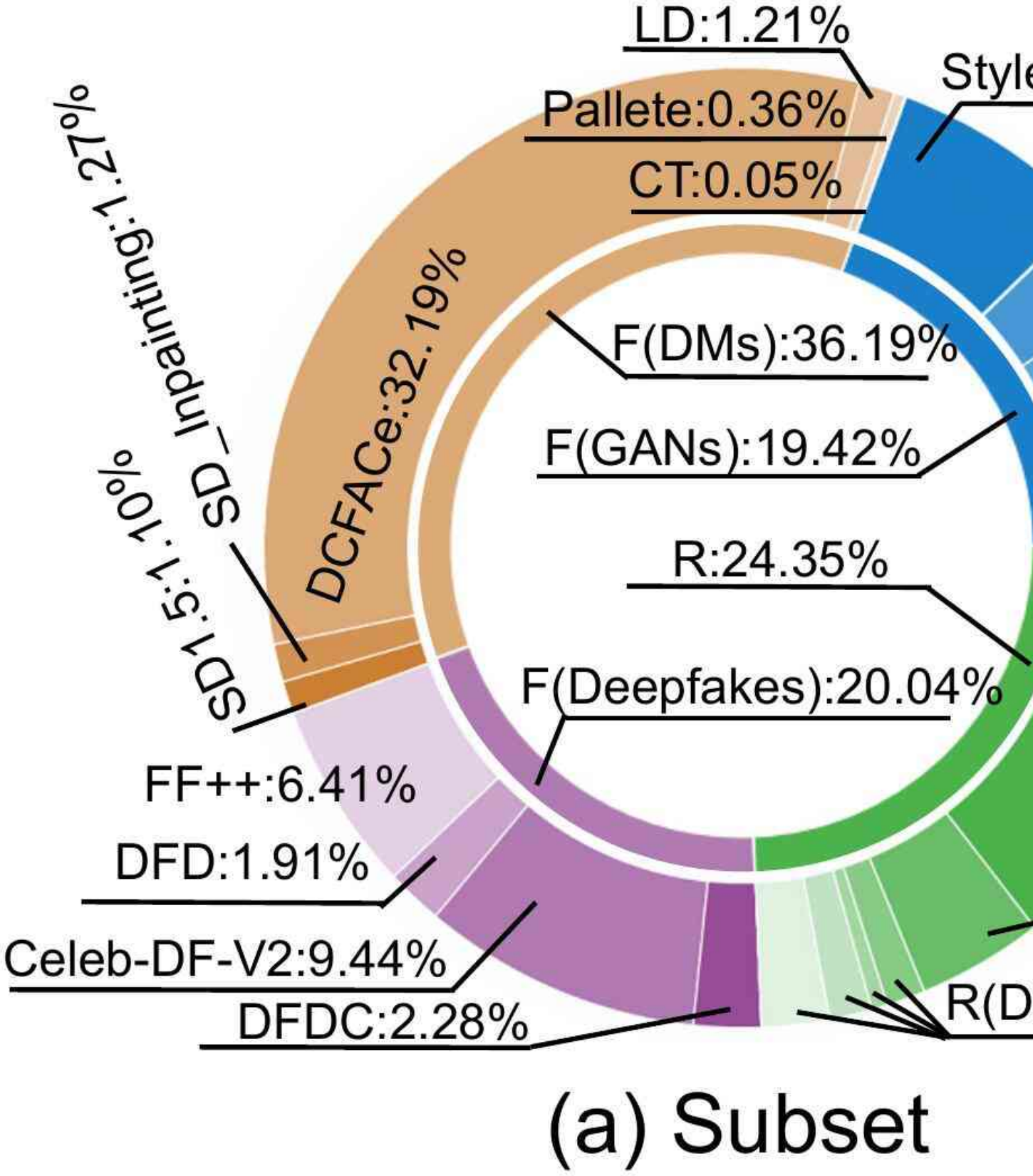

Figure 2. Distributional coverage across domains and difficulty bands.

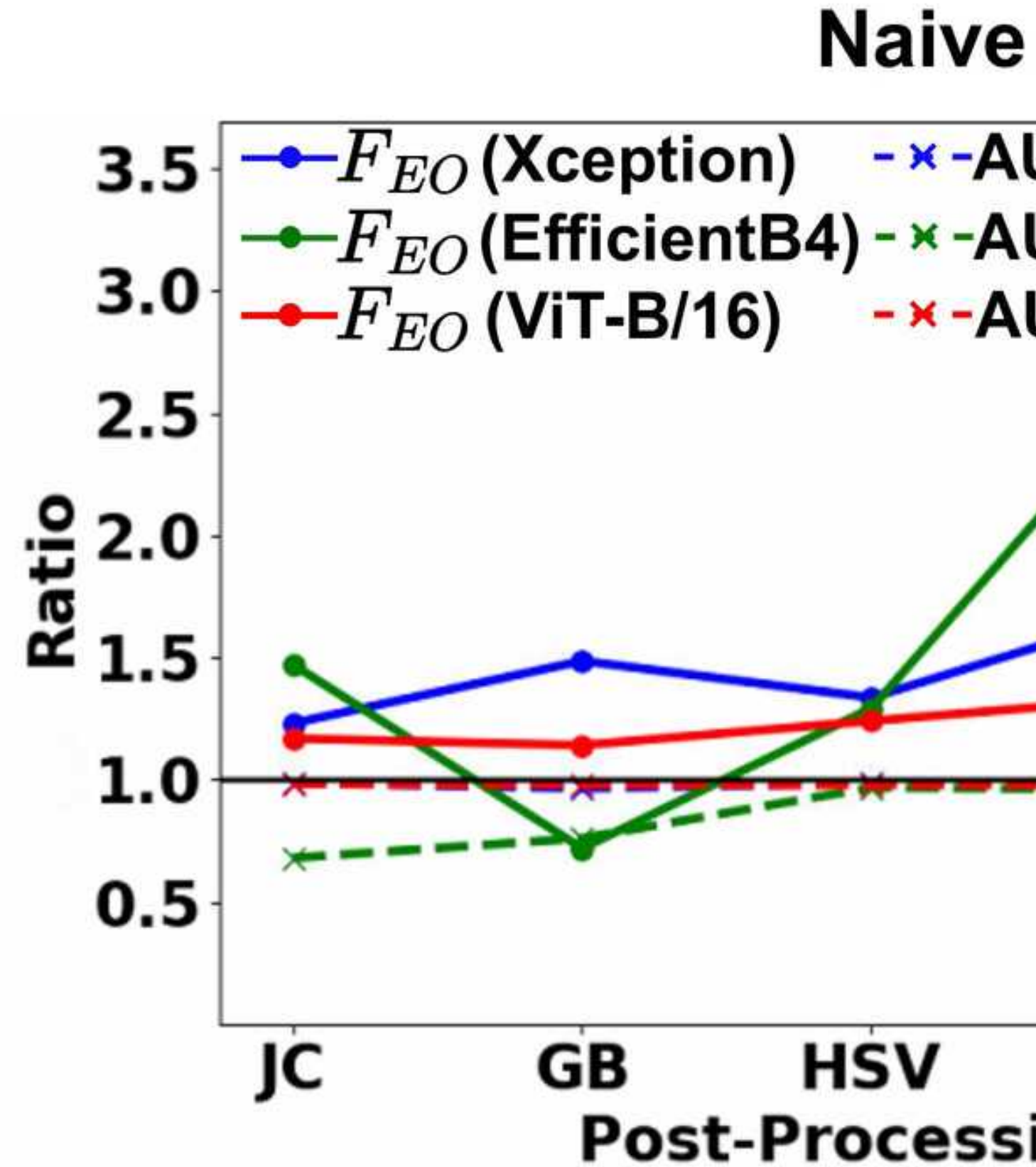

Figure 3. Robustness under controlled perturbations; error bars show 95% CIs.